EXTREME PROGRAMMING(XP)

Extreme Programming (XP) is one of the numerous Agile frameworks applied by IT companies. But its key feature — emphasis on technical aspects of software development — distinguishes XP from the other approaches.

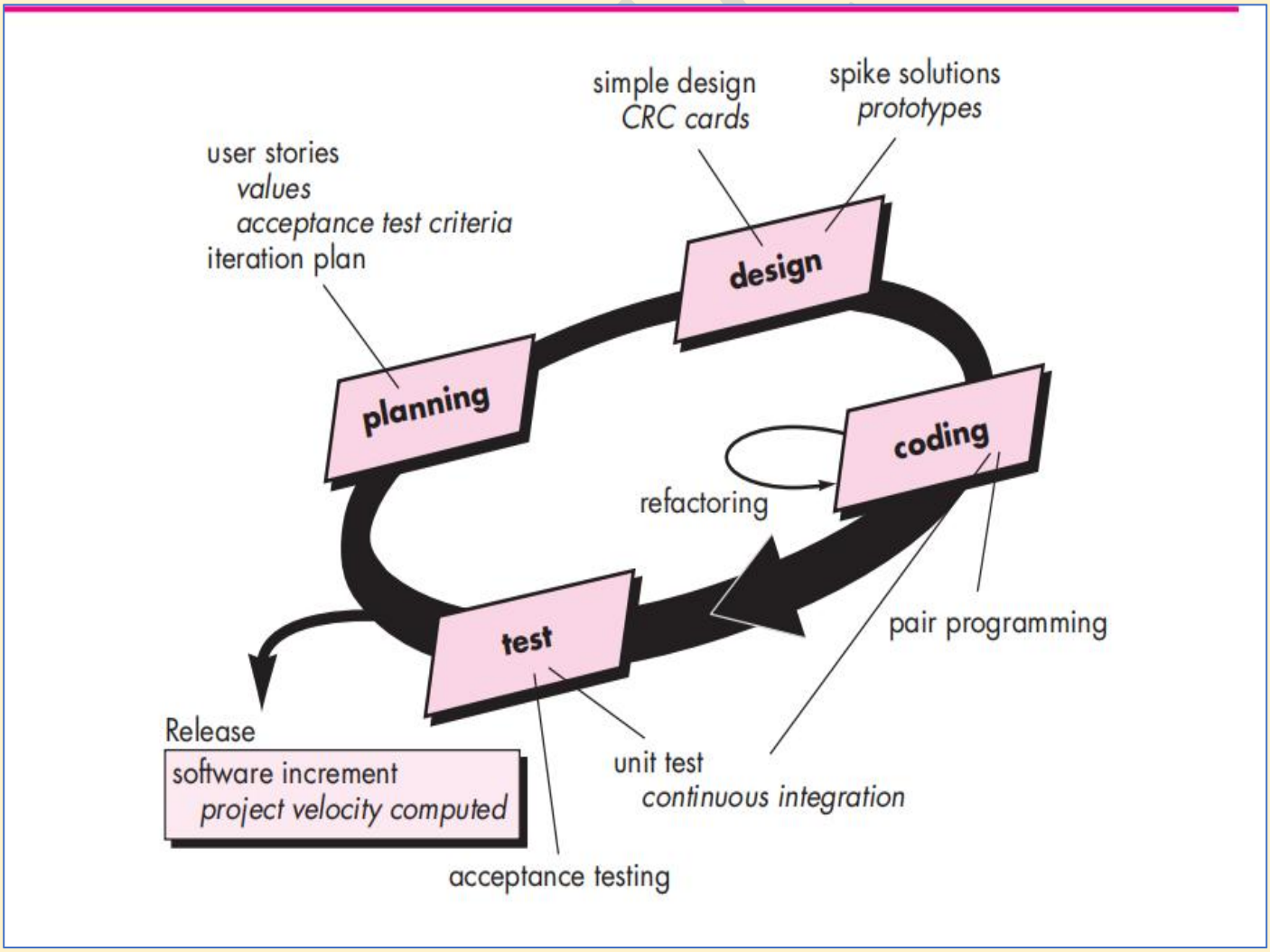
• XP is a lightweight, efficient, low-risk, flexible, predictable, scientific, and fun way to develop a software.

• Small to medium sized teams that work under vague and rapidly changing requirements.

• The five values of XP are communication, simplicity, feedback, courage, and respect

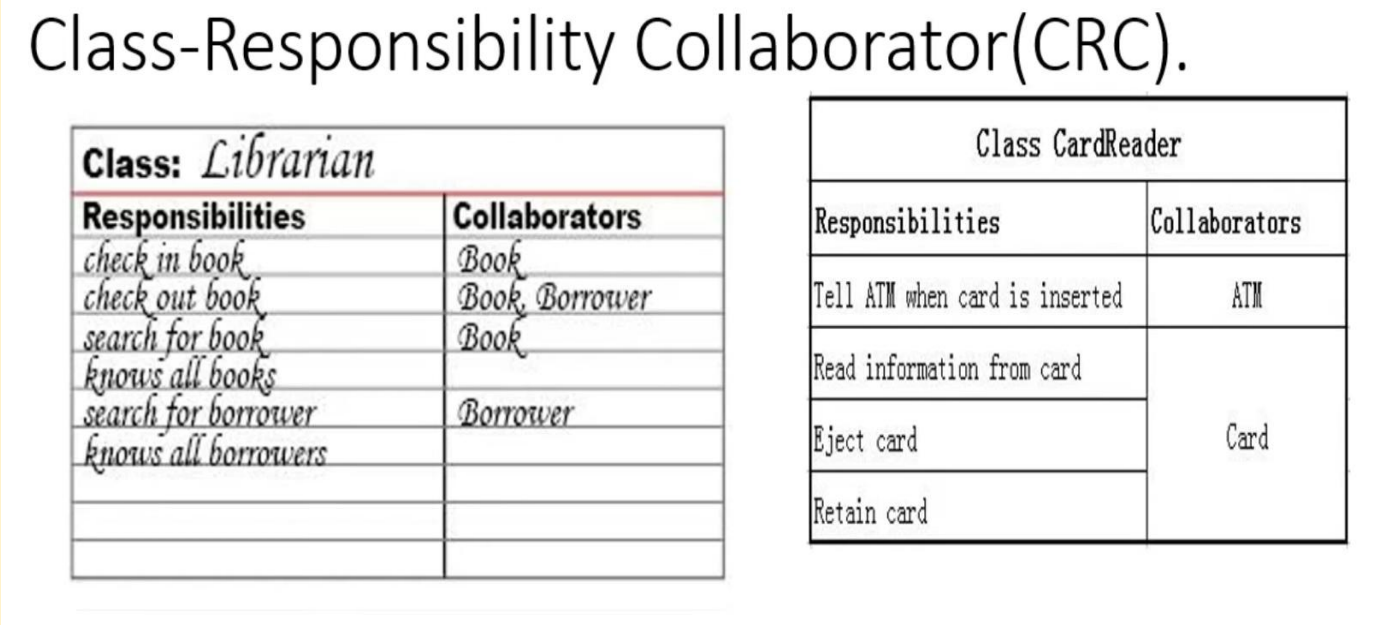
• Follows Object Oriented approach.

The Process & Roles of Extreme Programming



>>Keywords:

• **CRC cards:** CRC cards, or Class-Responsibility-Collaboration cards, are a brainstorming and design tool used in object-oriented software engineering. They consist of index cards divided into three sections, which help in identifying the classes within a system, along with their responsibilities and collaborators.



• **Spike solution Prototype:** A spike solution is a short-term experiment that helps developers explore complex or unknown issues. They can be useful for testing feasibility, reducing risks, and learning new skills or technologies.

• **Refactoring (Simplify):** Refactoring is about removing redundancy, eliminating unnecessary functions, increasing code coherency, and at the same time decoupling elements. *Keep your code clean and simple*, so you can easily understand and modify it when required would be the advice of any XP team member.

• **Pair Programming:** This practice requires two programmers to work jointly on the same code.

• **Project Velocity:** It is a metric used to measure how much work a team can complete within a specific time frame.

Values of extreme programming

XP has simple rules that are based on 5 values to guide the teamwork:

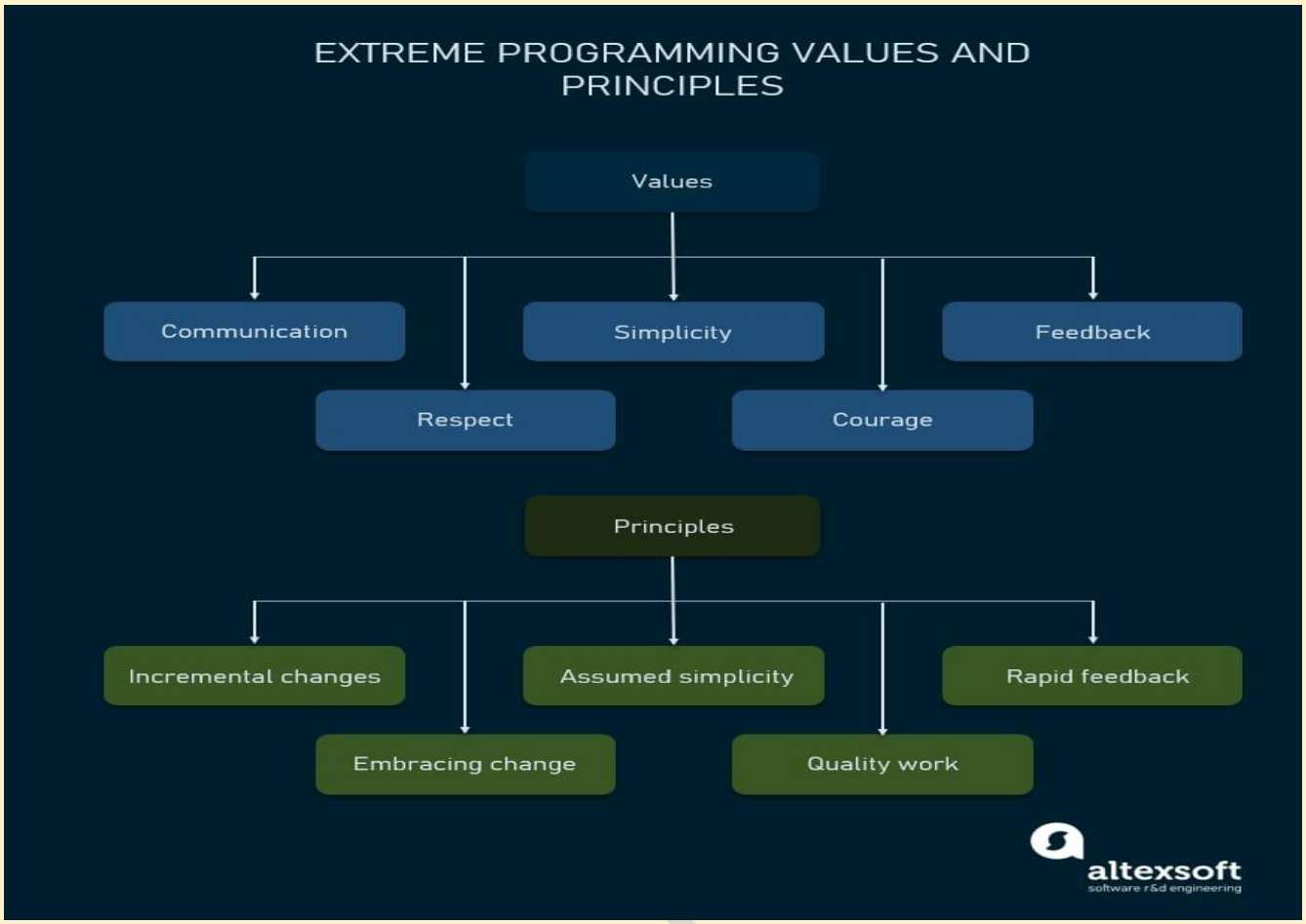
1. **Communication.** Everyone on a team works jointly at every stage of the project.
2. **Simplicity.** Developers strive to write simple code bringing more value to a product, as it saves time and effort.
3. **Feedback.** Team members deliver software frequently, get feedback about it, and improve a product according to the new requirements.
4. **Respect.** Every person assigned to a project contributes to a common goal.
5. **Courage.** Programmers objectively evaluate their own results without making excuses and are always ready to respond to changes.

These values represent a specific mindset of motivated team players who do their best on the way to achieving a common goal. XP principles derive from these values and reflect them in more concrete ways.

Principles of extreme programming

Most researchers denote 5 XP principles as:

1. **Rapid feedback.** Team members understand the given feedback and react to it right away.
2. **Assumed simplicity.** Developers need to focus on the job that is important at the moment and follow YAGNI (You Ain't Gonna Need It) and DRY (Don't Repeat Yourself) principles.
3. **Incremental changes.** Small changes made to a product step by step work better than big ones made at once.
4. **Embracing change.** If a client thinks a product needs to be changed, programmers should support this decision and plan how to implement new requirements.
5. **Quality work.** A team that works well, makes a valuable product and feels proud of it.



EXTREME PROGRAMMING(XP)

- **ADVANTAGES:-**

1. Fewer documentation required.
2. Collaboration with customers.
3. Flexibility to developers.
4. Easy to manage.

- **DISADVANTAGES:-**

1. Depends heavily on customer interaction.
2. Transfer of technology to new team members may be quite challenging due to lack of documentation.

WEBSITE LINK- https://www.altexsoft.com/blog/extreme-programming-values-principles-and-practices/

Video Link- https://www.youtube.com/watch?v=7G4uN4S0jD4

Lean Software Development(LSD)

The **Poppendiecks** created **Lean Software Development (LSD)** by importing lean manufacturing ideas into software development. **LSD** focuses more on principles and, to me, feels less concrete compared to other agile methods.

The Process Of LSD:

Now look at these 7 Core Concepts in more detail:

1. Eliminate Wastes: To maximize value, We must minimize Waste. For software systems, Waste can take the form of partially done work, delays, hand-offs, unnecessary features etc.. Therefore to increase the value we are getting from projects, we must develop ways to identify and then remove waste.

Example of wastes-

No	Waste
1	Partially done work
2	Extra Processes
3	Extra Features
4	Task switching
5	Waiting
6	Motion
7	Defects

No	Waste	Description	Example
1	Partially done work	Work started, but not complete;	➢ Code waiting for testing ➢ Specs waiting for development
2	Extra Processes	Extra work but does not add value	➢ Unused documentation ➢ Unnecessary approvals
3	Extra Features	Features that are not required, or thought of as "nice-to-have" features	➢ Gold Plating ➢ Technology Features

No	Waste	Description	Example
4	Task switching	Multi tasking between several different projects when there are context-switching penalties	People on multiple projects
5	Waiting	Delays waiting for reviews and approvals	➢ Waiting for prototype reviews ➢ Waiting for document approvals

No	Waste	Description	Example
6	Motion	The effort required to communicate or move information or deliverables from one group to another	➢ Distributed teams ➢ Handoffs
7	Defects	Defective documents or software that need correction	➢ Requirements defects ➢ Software bugs

2. Empower the team: Rather than taking a micromanagement approach, we should respect team members superior knowledge of the technical steps required on the project and let them make local decisions to be productive and successful.

3. Deliver Fast: We can maximize the project Return on investment (ROI) by quickly delivering valuable software and iterating through designs. We find the best solution through the Rapid Evolution of options. (It basically follows incremental model)

4. Optimize the Whole: We aim to see the system as more than the sum of its parts. We go beyond the pieces of the project and look for how it aligns with the organization. As part of optimizing the whole, we also focus on forming better inter-group relations.

5. Build quality in: Lean development doesn't try to "test-in" quality at the end; instead, we build quality into the product and continually assure quality throughout

the development process, using techniques like refactoring, continuous integration and unit testing. (Quality over Quantity)

6. Defer Decisions: We balance early planning with making decisions and committing to things as late as possible. For example, this may mean re-prioritizing the backlog right up until it is time to plan an iteration, or avoiding being tide to an early technology-bounded solution.

Example- for example in 'US' they will not allow to choose the major stream at first like AI/DataScience/Machine Learning/Cyber security/Game-development/graphic design/web developmet/app development etc. Only they will focus on Common stream like CSE for first 2 or 3 years in B.Tech/B.E, after 2 or 3 years student will decide & get into their own Interested domain.

7. Amplify Learning: This concept involves facilitating communication early and often, getting feedback as soon as possible, and building on what we learn. Software projects are business and technology learning experiences, so we should start soon and keep learning. (It means we need to learn from the ground or base & we need to listen feedback from team members in both ways one is the Content point of view and another is the emotional point of view.)

WEBSITE LINK- https://hangoutagile.com/lean-software-development-wave-ii/

Video Link- https://www.youtube.com/watch?v=aoHDa3L2ngI

GitHub

>> What Is GitHub?

GitHub is an online platform for software development that allows users to store, track, and collaborate on software projects. It utilizes Git, an open-source version control system, enabling multiple developers to work on a project simultaneously.

>> Why We Use GitHub?

GitHub is used for storing, tracking, and collaborating on software projects. It allows developers to maintain code in repositories, manage version control, and work together with others on coding tasks, which enhances collaboration and project management. (mainly publish our project & for group work also)

>> What is Version Control?

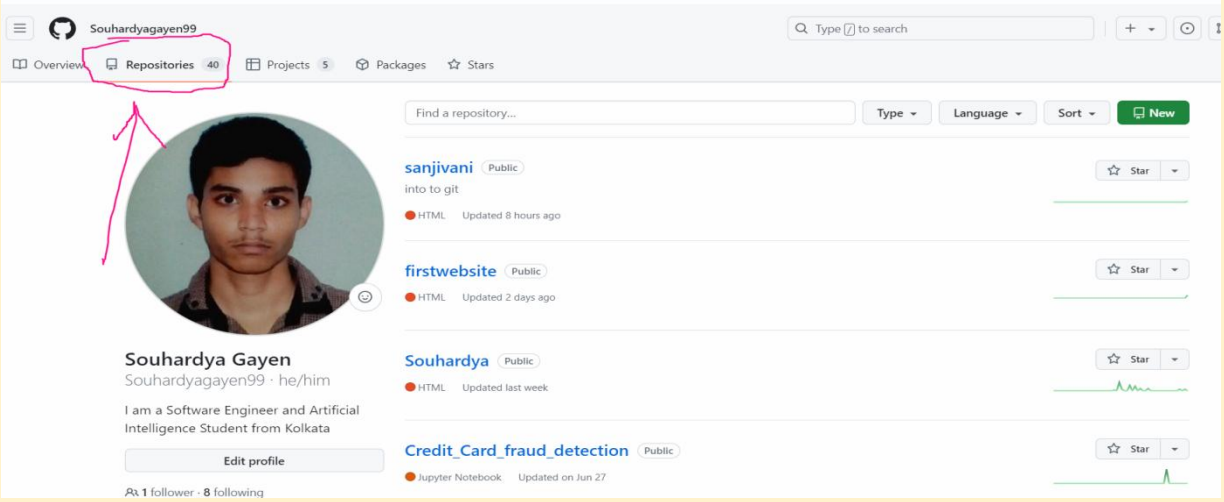
Version control is a system that helps manage changes to documents, programs, and other collections of information over time. It allows multiple users to collaborate on projects while keeping track of each change made. Here are some key aspects of version control. (for updating a version of any application/website/software)

>> How does version control work in GitHub?

Version control in GitHub is primarily managed through Git, a distributed version control system that allows multiple people to work on a project simultaneously without overwriting each other's changes.

>> What is a GitHub repository?

It's a place where you can store your code, your files, and each file's revision history. Repositories can have multiple collaborators and can be either public, internal, or private.

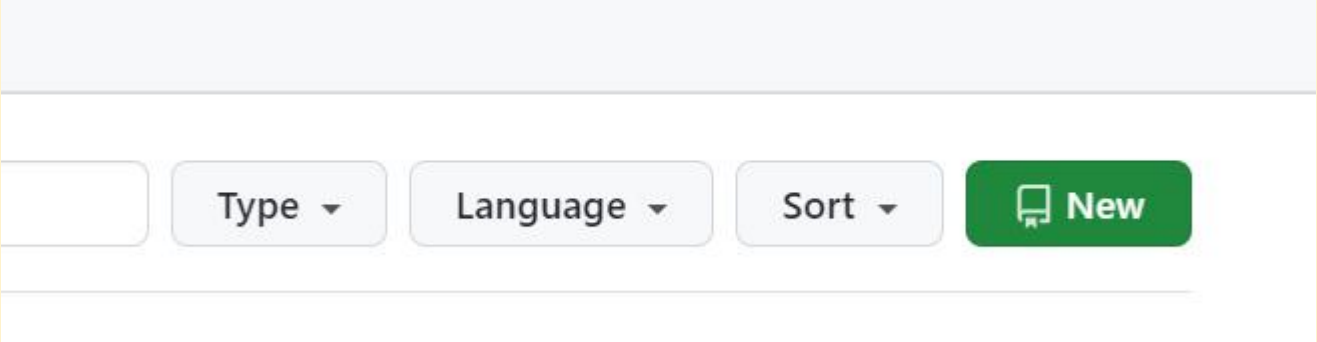


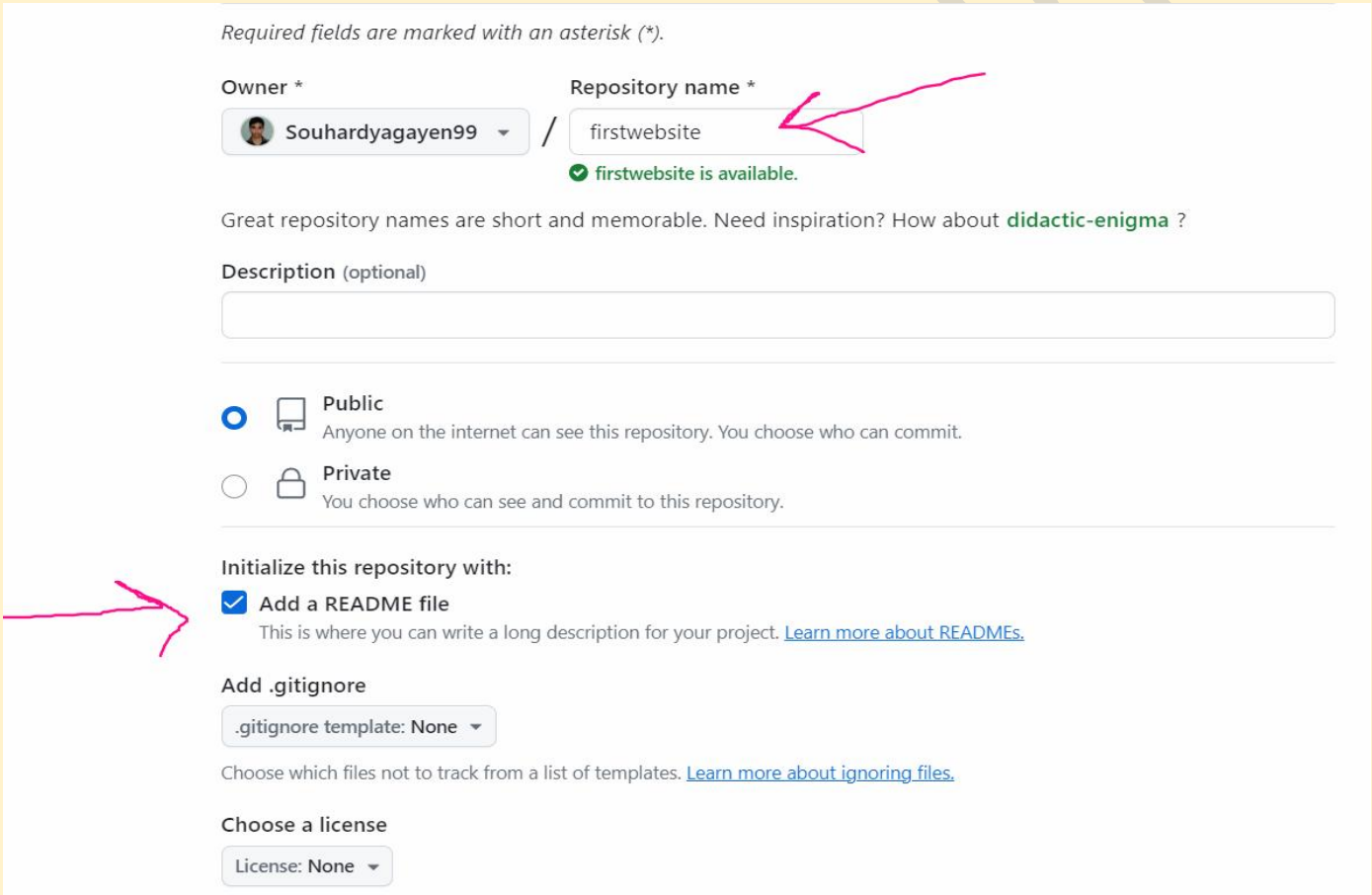
DO NOT COPY

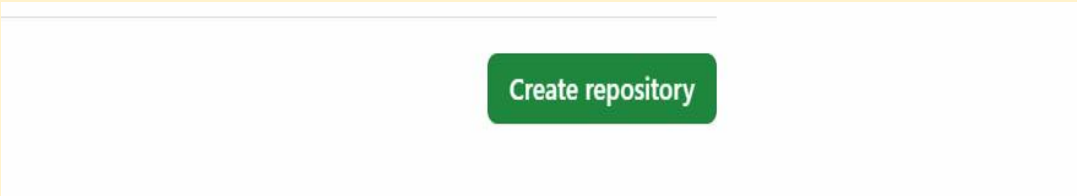
First of all, Create an ID on GitHub [means Sign Up and Sign In]

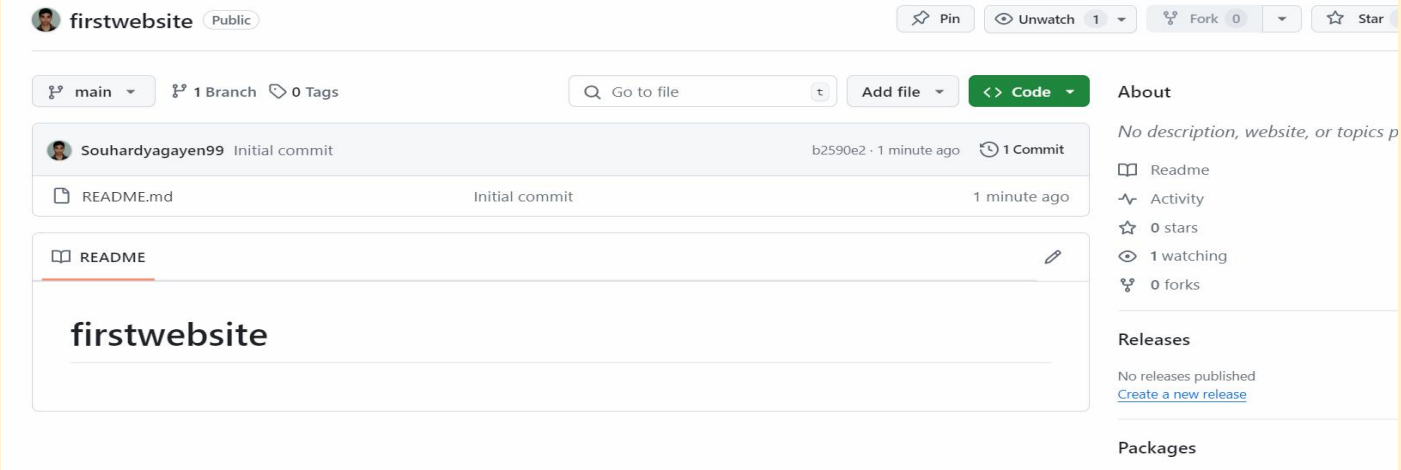
(link- https://github.com/)

After that Create GitHub "New Repositories"









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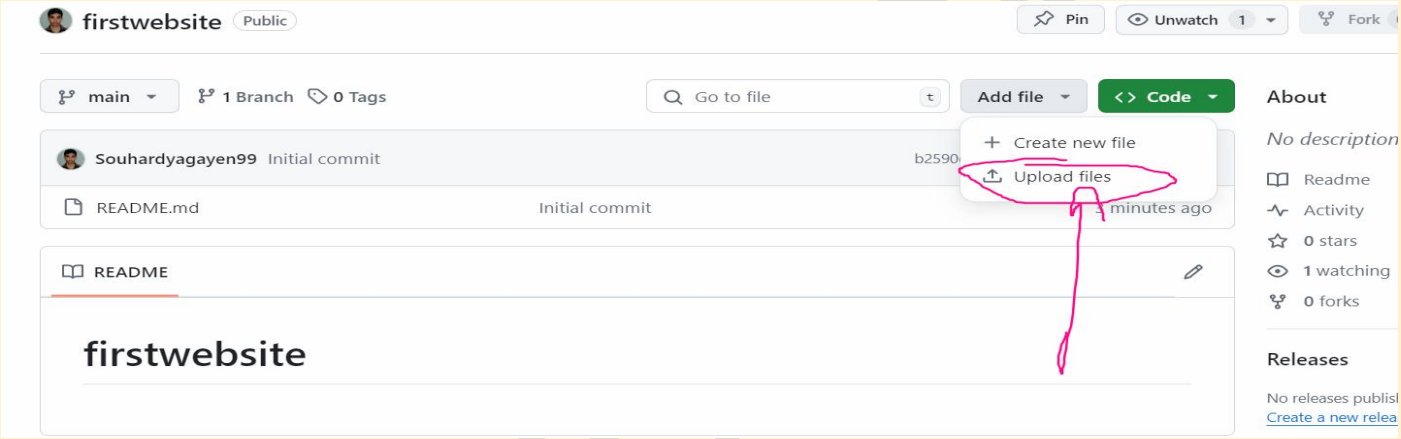
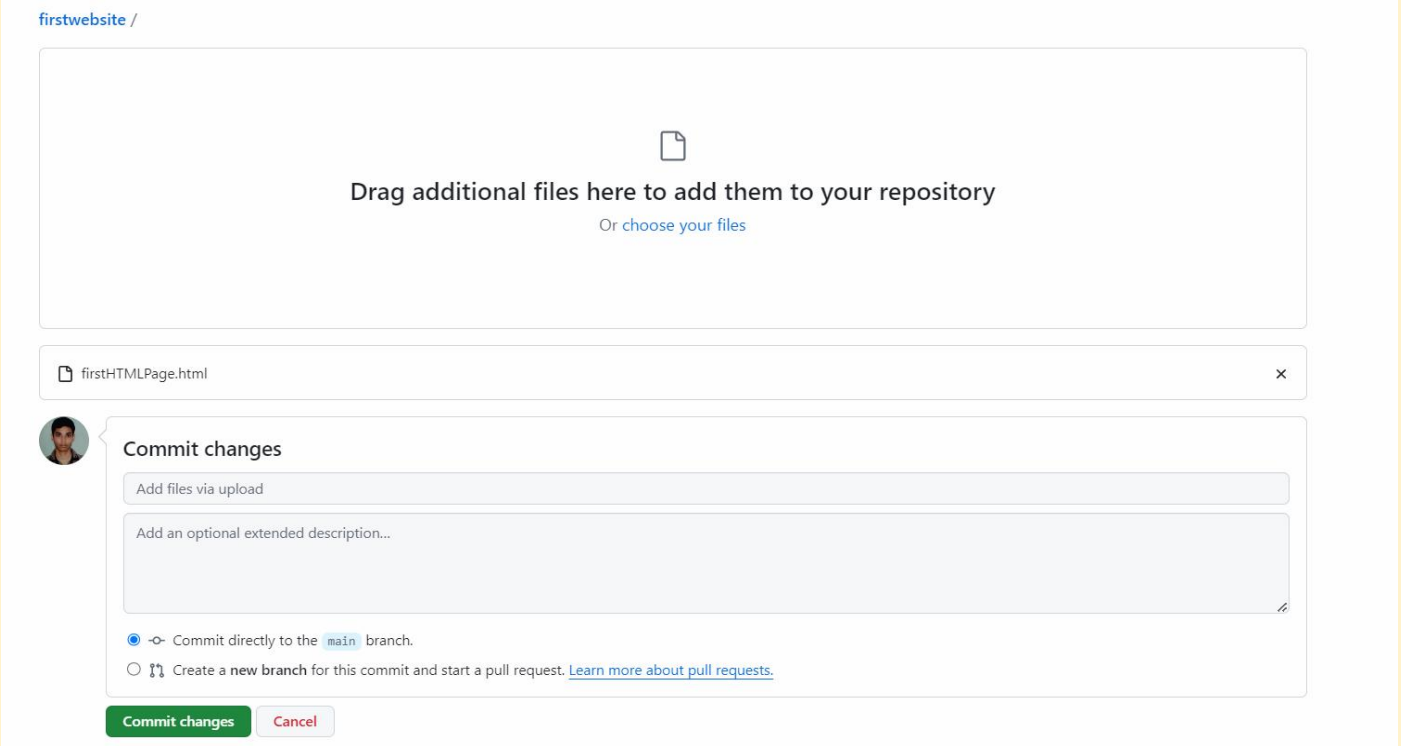
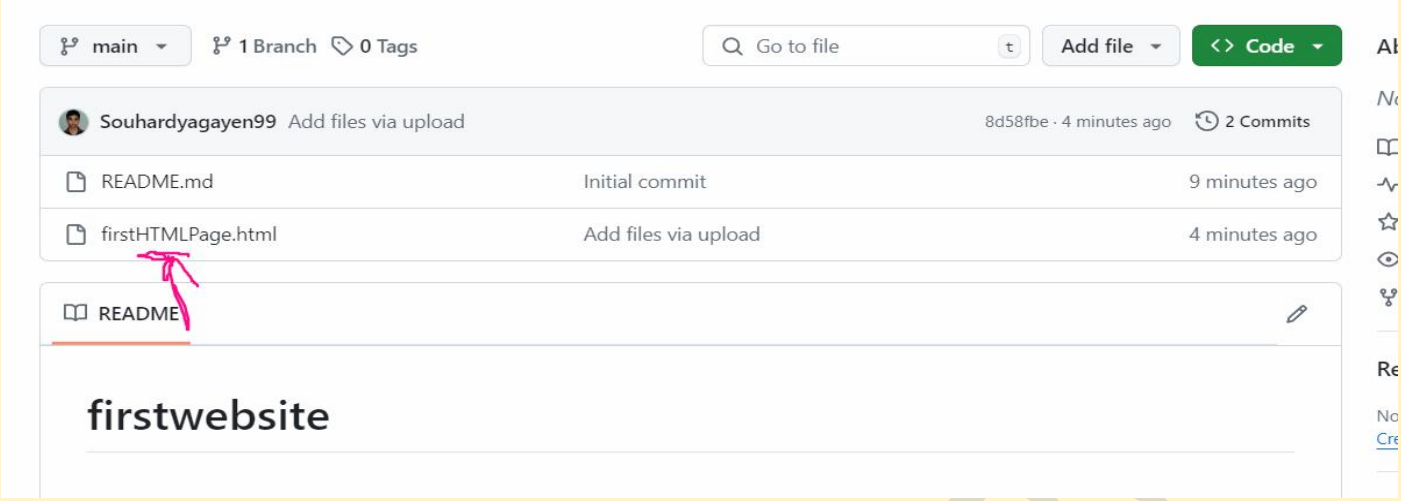
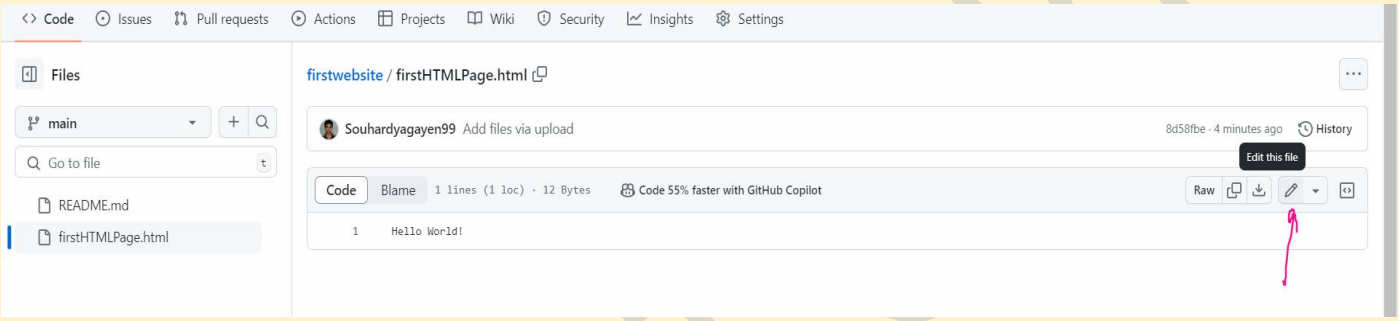


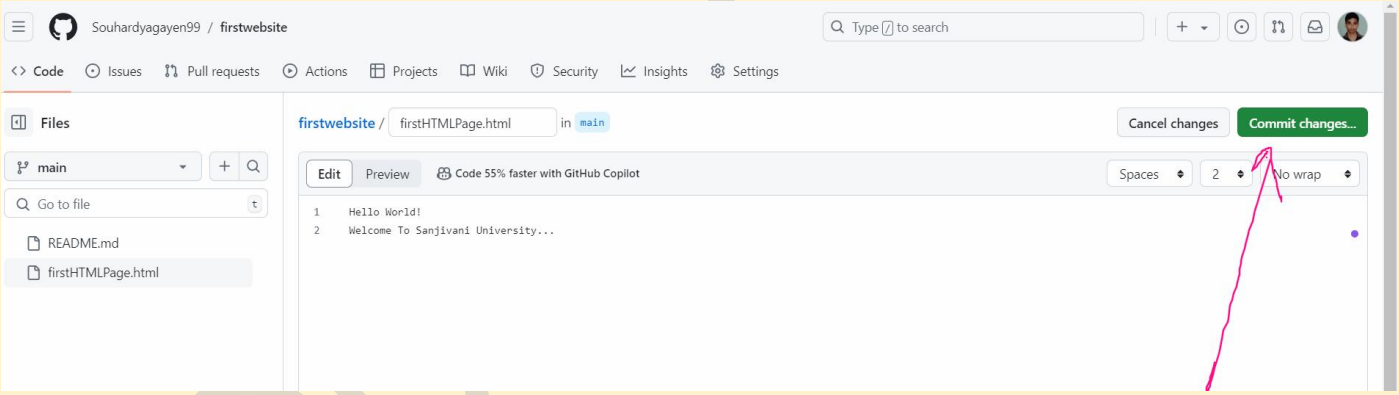


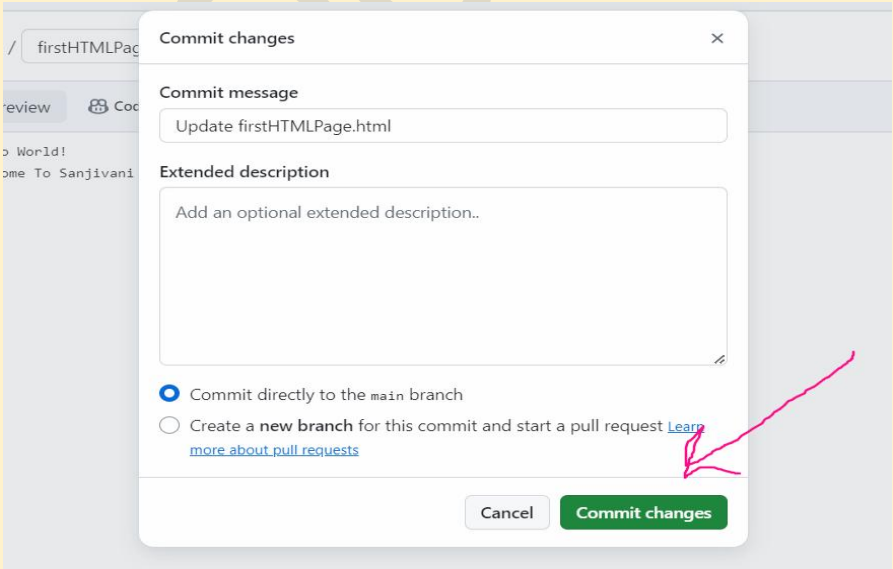
After Upload click “Commit Changes”

>>Then EDIT File

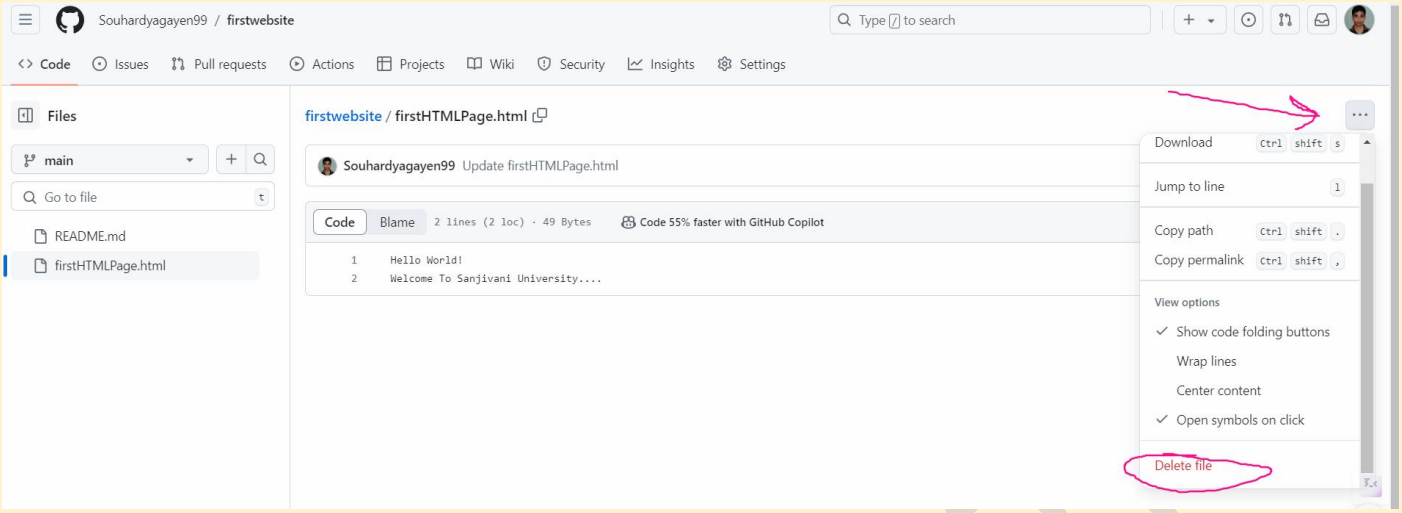


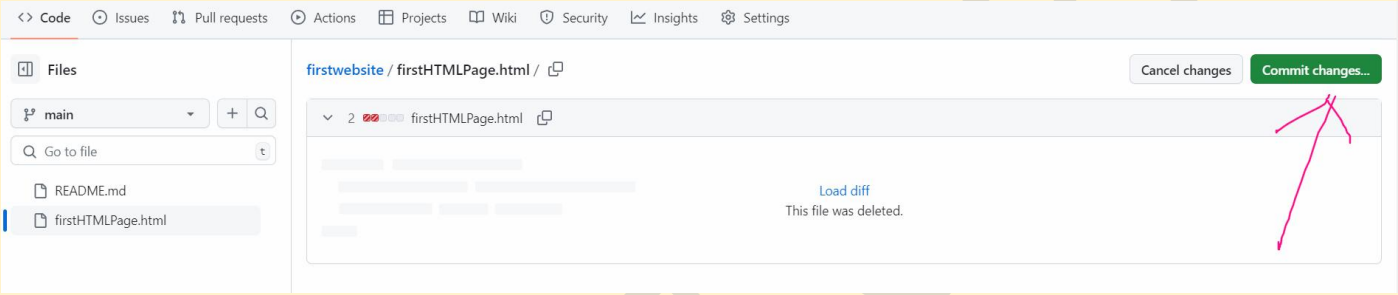


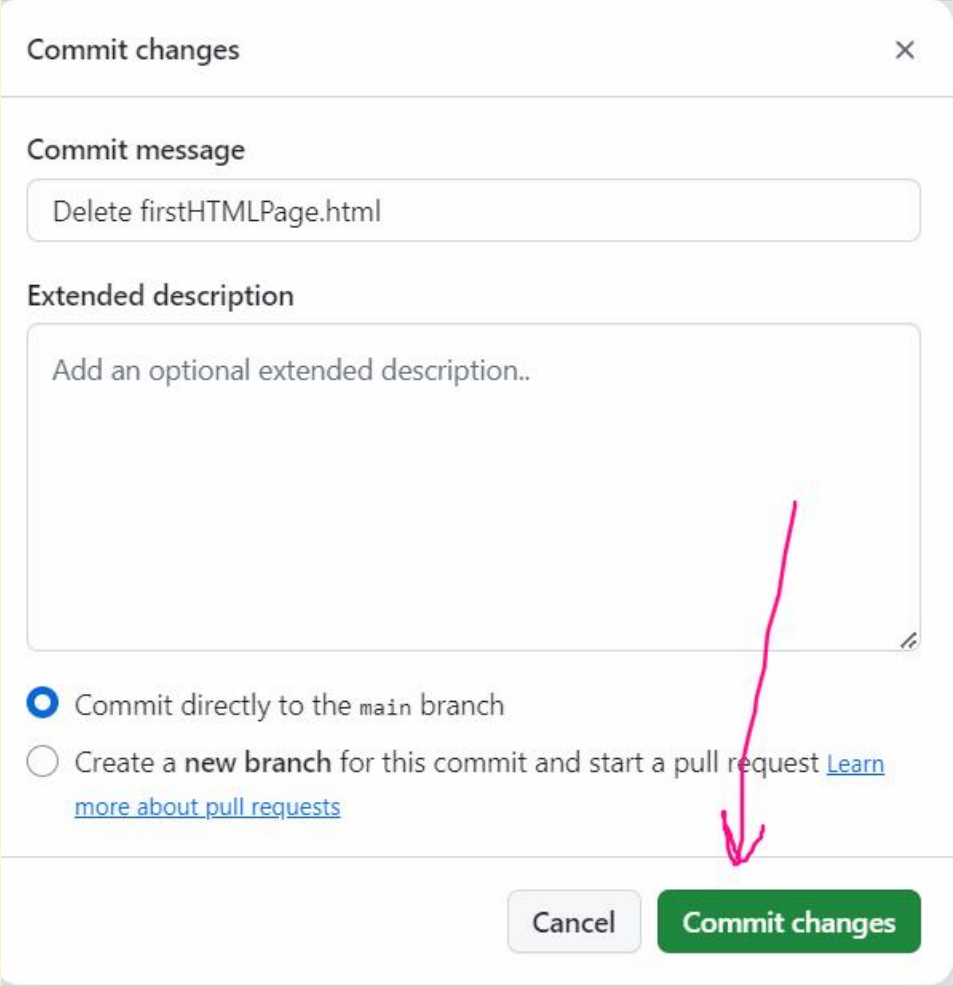




>>There-After Delete a Particular file

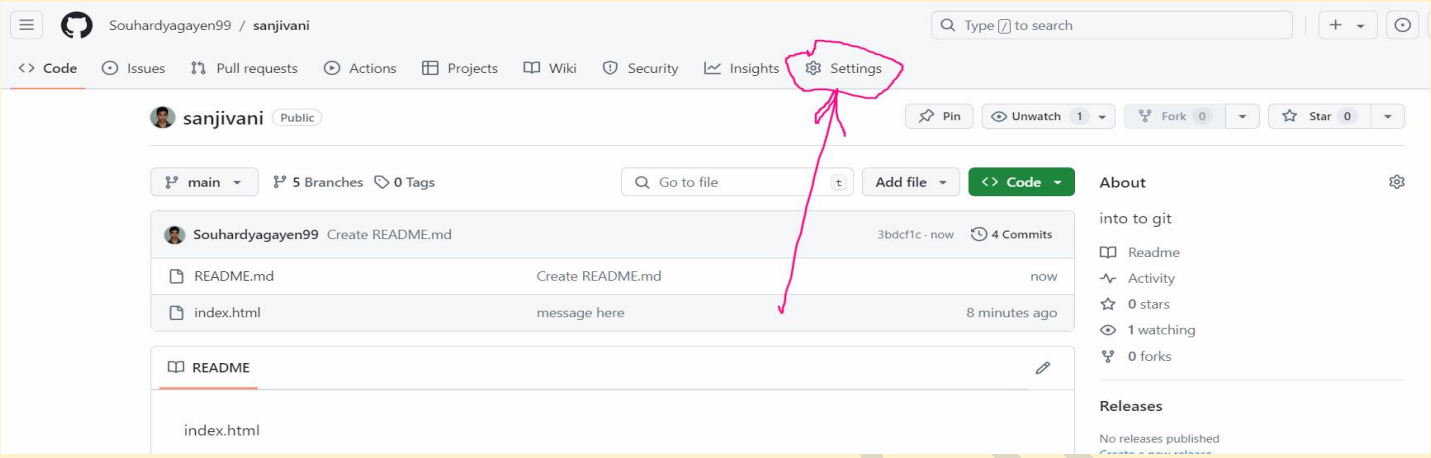


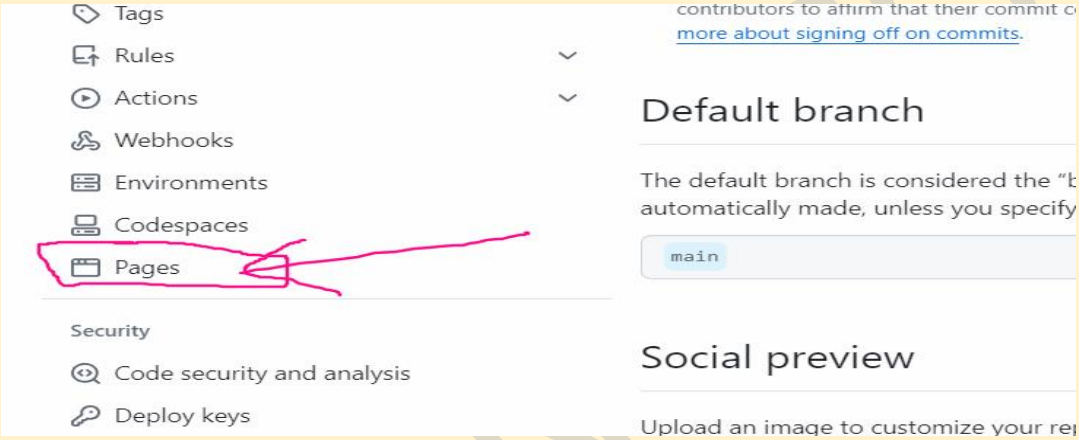


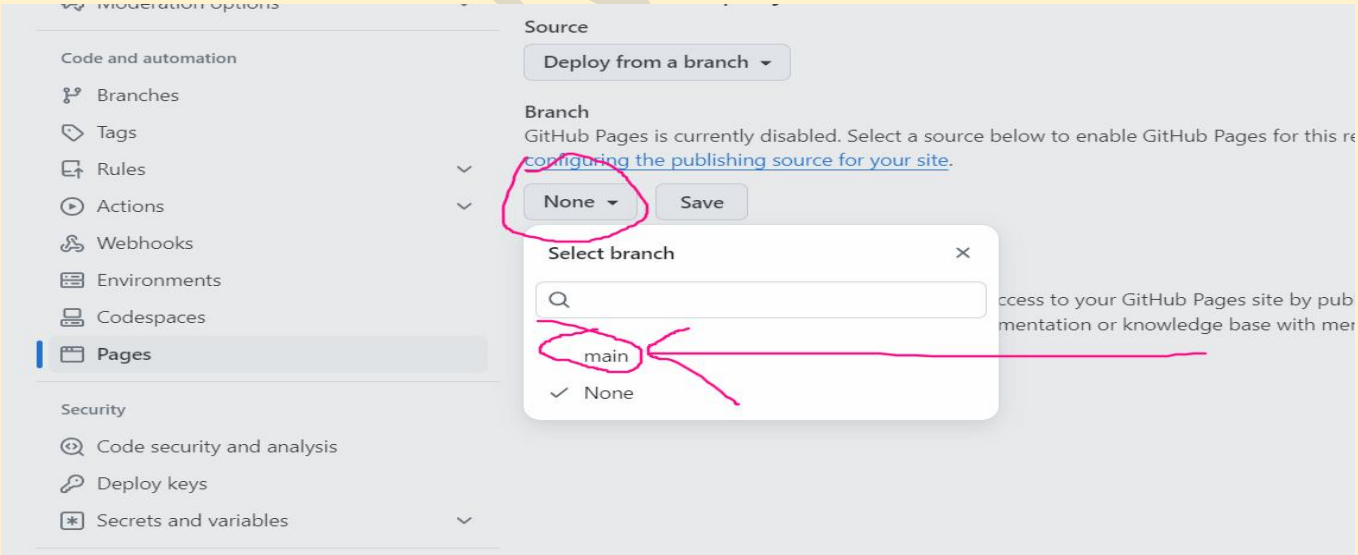


>>To Deploy/Publish/Host Our website

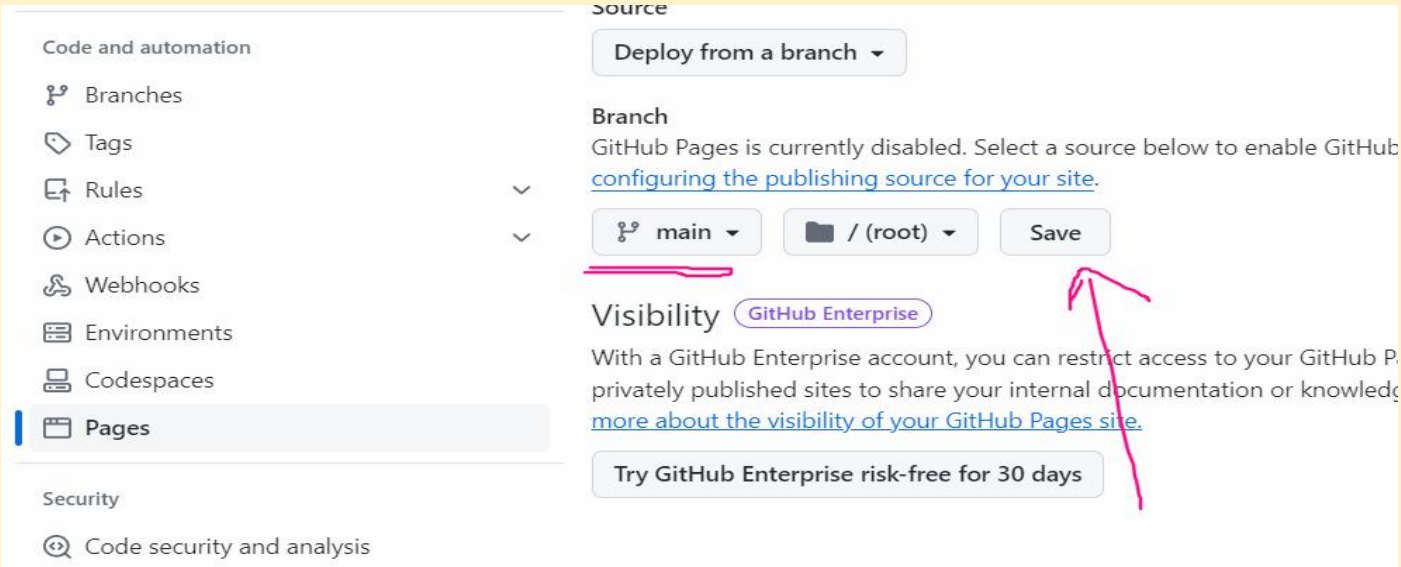
NOTE: For host in GitHub the name should be index.html otherwise it will not work Because github set this policy that Entry Page/Home Page/Landing Page Should be index.html



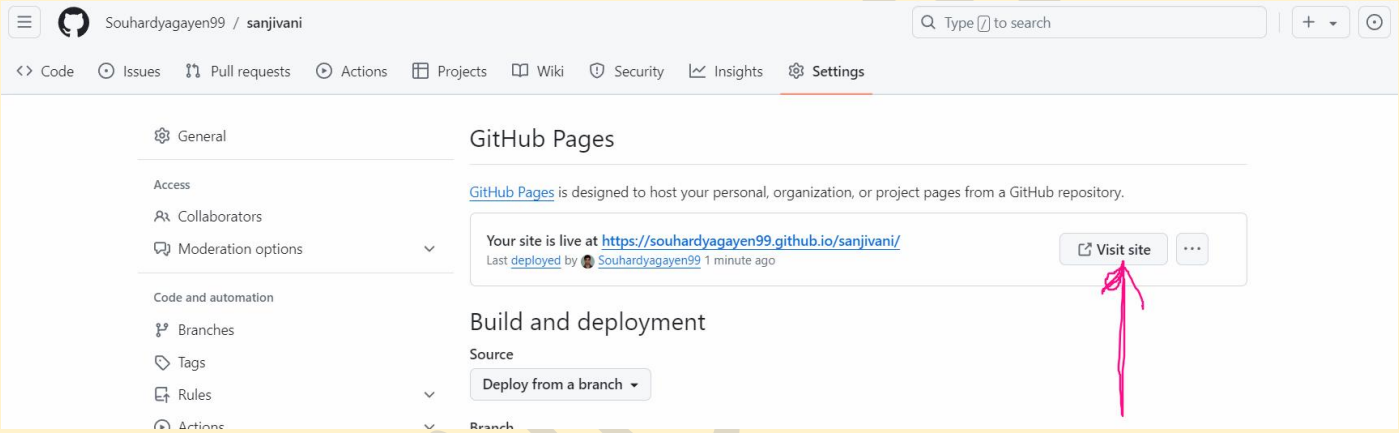




LINK- https://www.youtube.com/watch?v=4eMnz8VbgyM&t=305s (see the timestamp at 2:30 min)

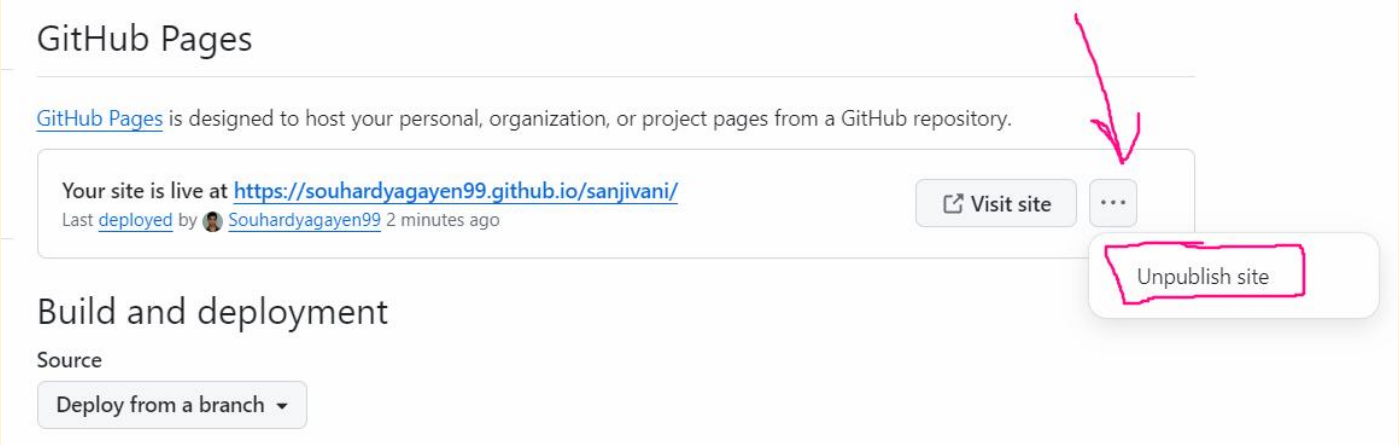


After 2-3 minutes Refresh/Reload the page and see…

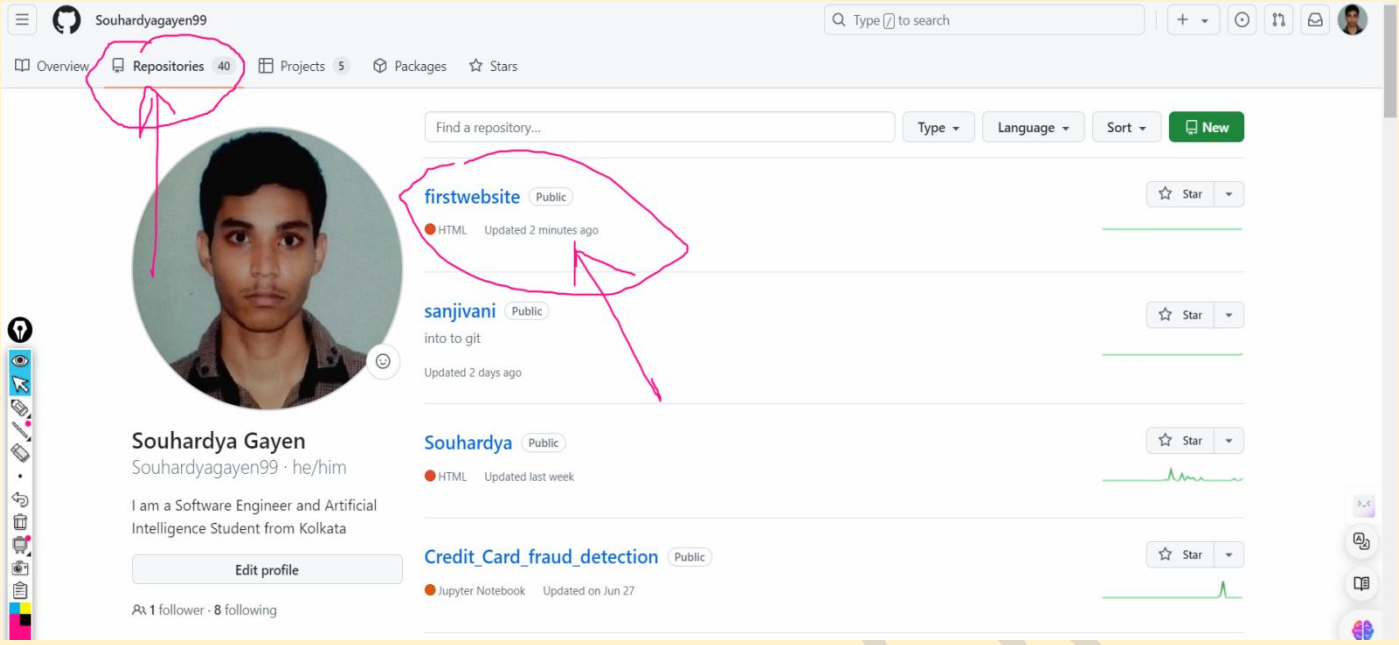


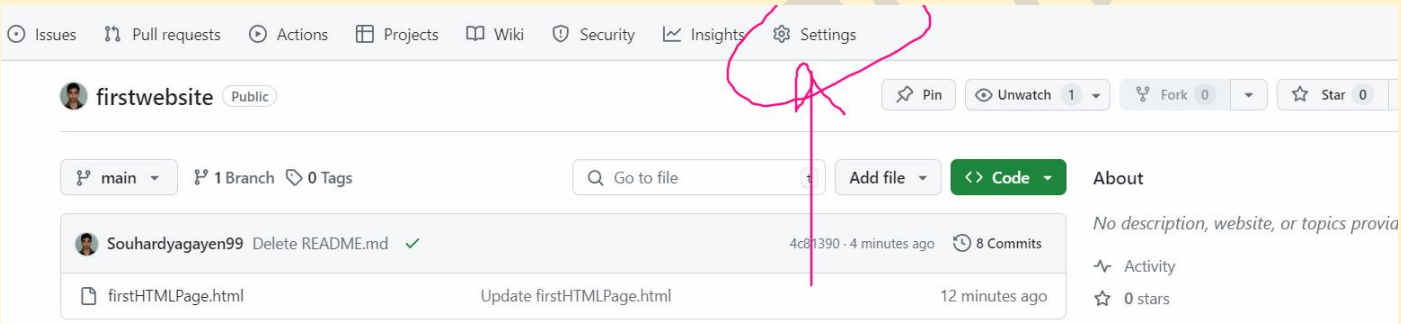
You can share this link with your friends.

For Unpublish.

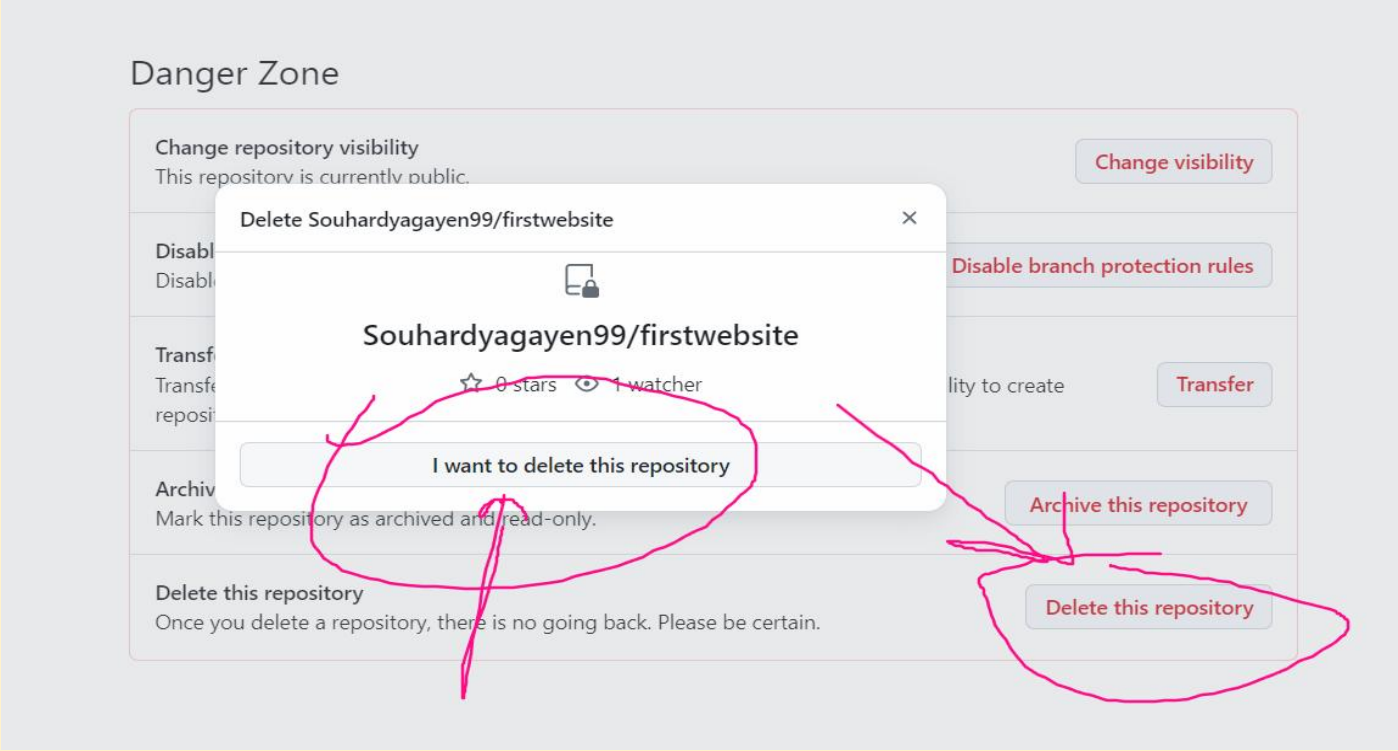


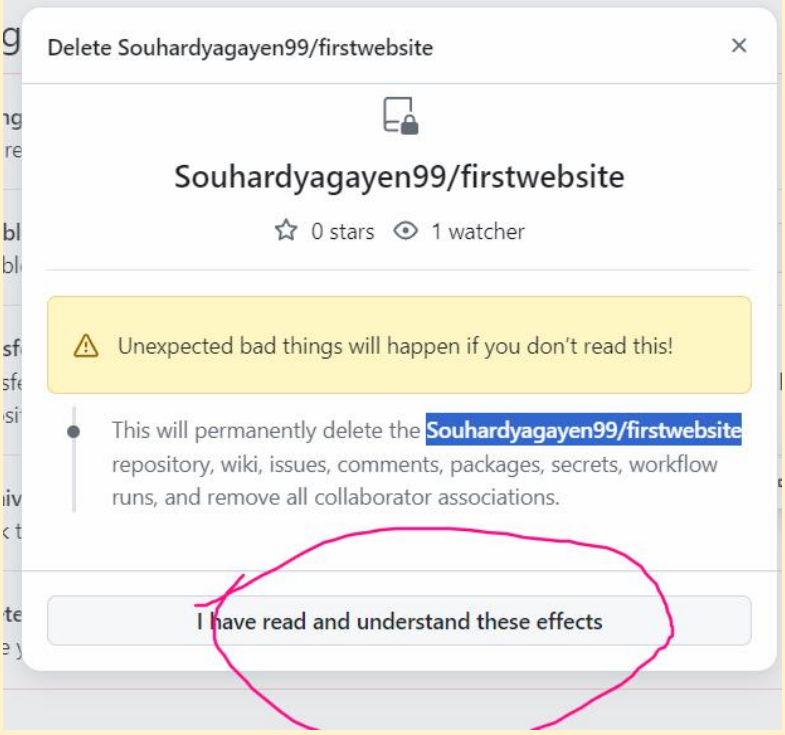
>>Now Delete A Full Repository

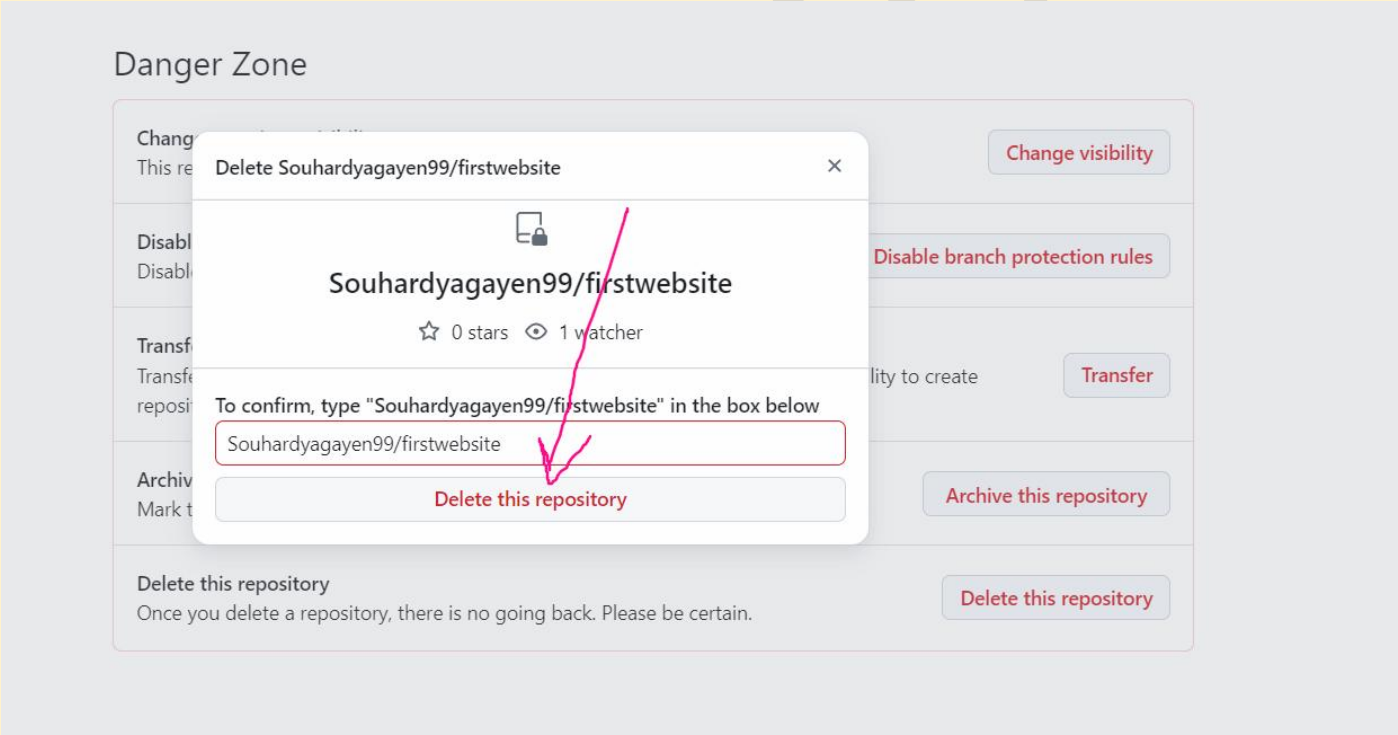




Scroll Down and go to bottom…




Delete Souhardyagayen99/firstwebsite
Souhardyagayen99/firstwebsite
0 stars
1 watcher
Unexpected bad things will happen if you don't read this!
This will permanently delete the Souhardyagayen99/firstwebsite repository, wiki, issues, comments, packages, secrets, workflow runs, and remove all collaborator associations.
I have read and understand these effects


Danger Zone
Change visibility
Disable branch protection rules
Transfer
Archive this repository
Delete this repository
Once you delete a repository, there is no going back. Please be certain.
Delete Souhardyagayen99/firstwebsite
Souhardyagayen99/firstwebsite
0 stars
1 watcher
To confirm, type "Souhardyagayen99/firstwebsite" in the box below
Souhardyagayen99/firstwebsite
Delete this repository

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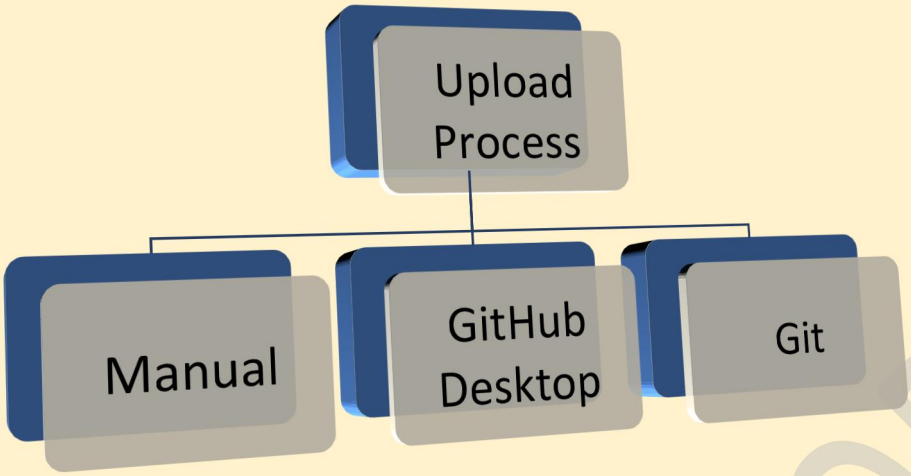
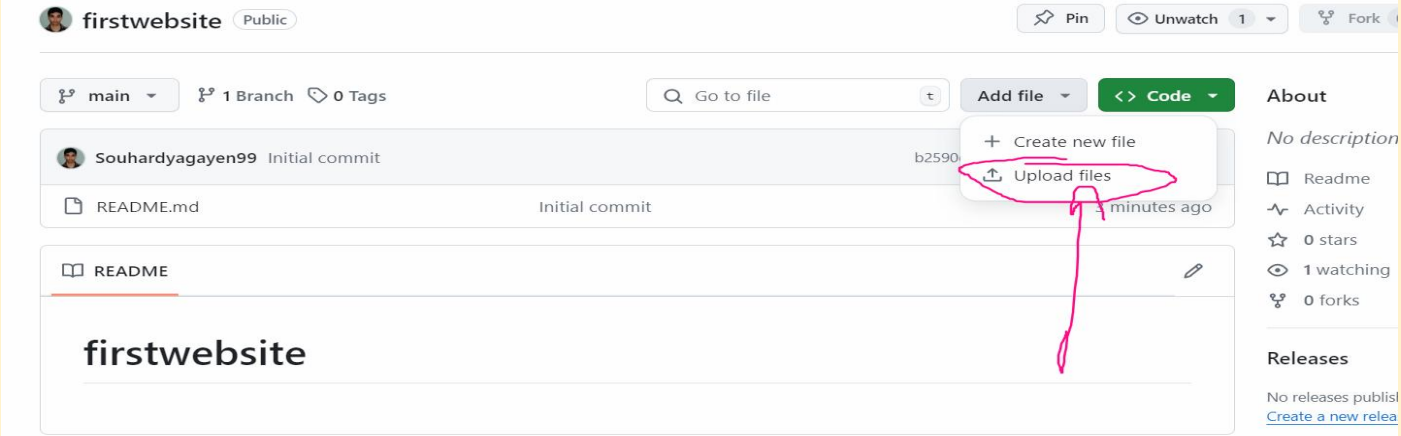
------→ P.T.O

>>TYPES OF UPLOADING FILE INTO GITHUB

- Upload file into GitHub Repositories through "Manual/Direct Drag & Drop Process" OR, "GitHub Desktop" OR, using "Git" you can upload.



Manual Process: It is a Drag & Drop Process It is actually for Small Projects.



Link-
https://www.youtube.com/watch?v=xmK1Q5uzH4w&list=PL0xmYJZV1DwDx8SlT70oug84M3PU1KEwb&index=13

GitHub Desktop: (It is Used for Large Projects)

At first, you have to create new repositories then add readme file then go to there where is your project file present.

Always Here We have to create New folder on Desktop & then clone the Repositories and then copy the code or file from your main original folder location.

Don't give the original file location where you have done your code otherwise it will replace your original files like readme...

Link-
https://www.youtube.com/watch?v=a28WGSNIEZc&list=PL0xmYJZV1DwDx8SlT70oug84M3PU1KEwb&index=8 (see that one)

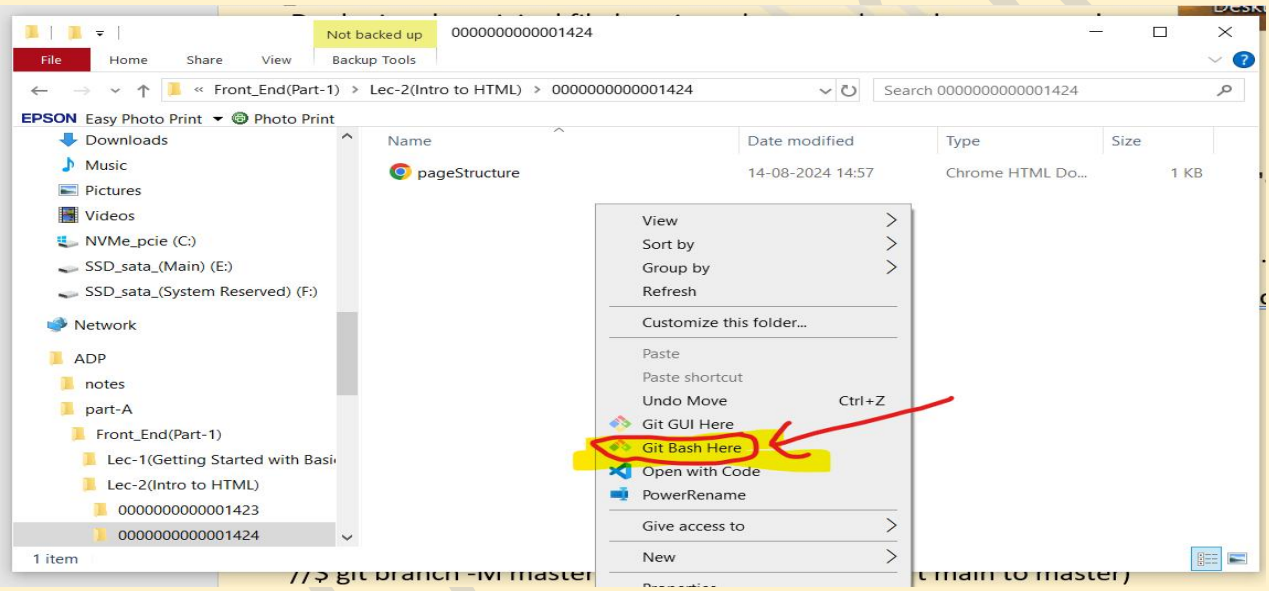
https://www.youtube.com/watch?v=eL7EOQ08-14&list=PL0xmYJZV1DwDx8SlT70oug84M3PU1KEwb&index=15

https://www.youtube.com/watch?v=A2R-CMMyp64&list=PL0xmYJZV1DwDx8SlT70oug84M3PU1KEwb&index=2&t=301s

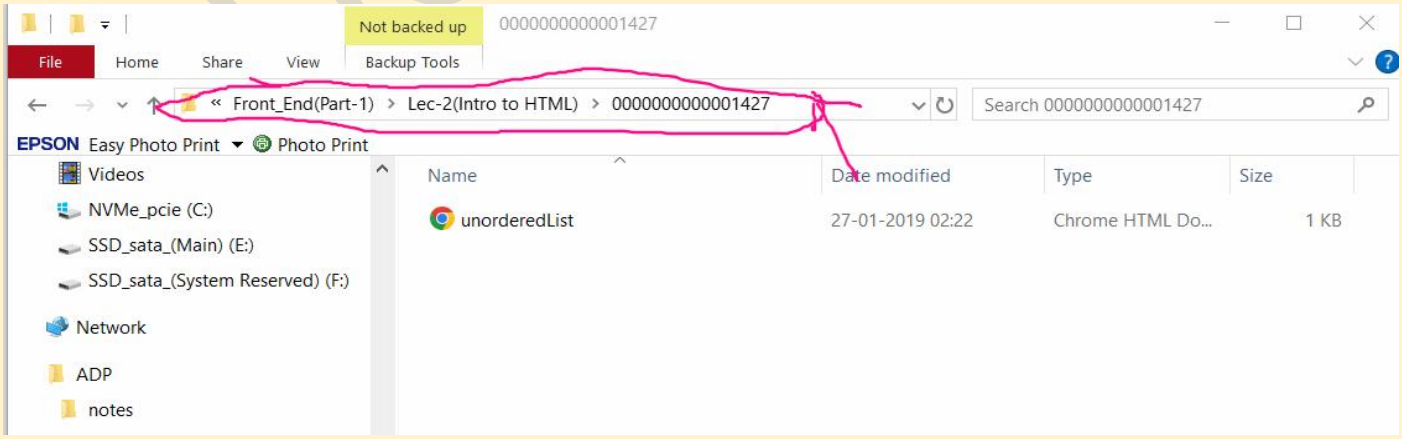
Git: (It is Used for Large Projects)

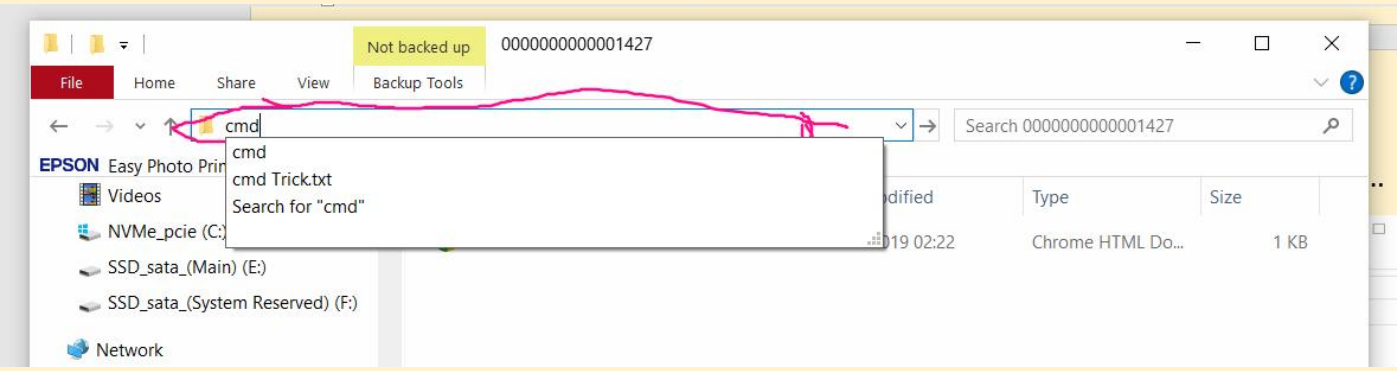
At first, you have to create new repositories then add readme file

Then go to there where is your project file present. Then right click and select "Git Bash here" and continue...

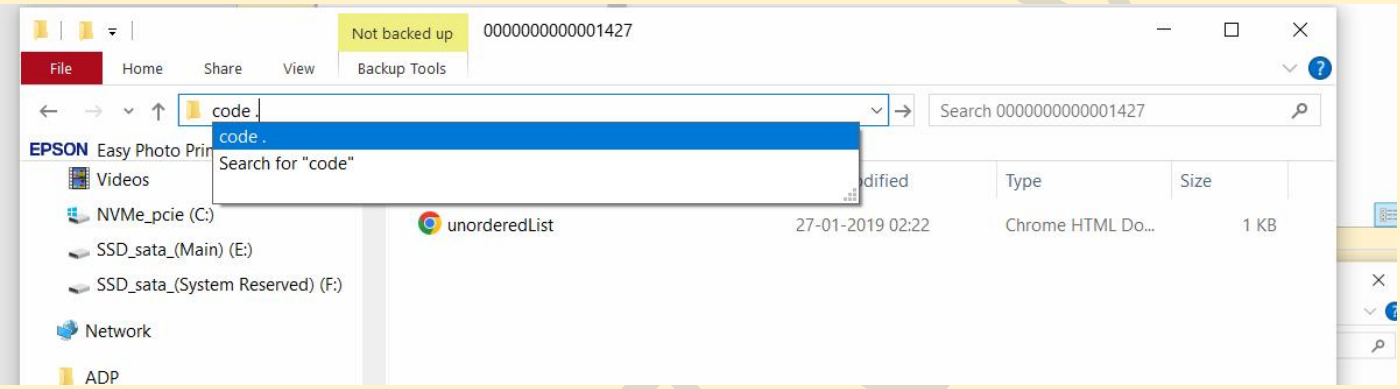


OR, select the address of your project file then type there “cmd” and continue…





OR, For VS Code(VS Code terminal) also select the address of your project file then type there “code .” and continue…



NOTE: Git Bash = CMD = VS Code Terminal (all are the same things)

<u>Now Execute This Commands…</u>

//$ echo "# ProjectChallenge" >> README1.md //(do not need) [it is for create README]

$ git init

$ git status //(to see status)

$ git branch -M main //you need to run(For convert master to main)

//$ git branch -M master //do not need(For convert main to master)

//$ git add README1.md // do not need(it is for add one file) OR

$ git add . // (to add all file)

$ git status

$ git commit -m "message here" // (Insteed of message here you can type any thing)

$ git remote add origin https://github.com/Souhardyagayen99/Online_Exam_System //(you need to copy the https code) OR

$ git remote add origin git@github.com:Souhardyagayen99/Online_Exam_System.git // (you need to copy the ssh code)

$ git status //(to see status)

//$ git push -u origin main //OR

//$ git push -u origin master //OR

//$ git push origin master //OR

//$ git push origin main //OR

//$ git push origin master --force //OR

$ git push origin main --force // (USE THAT FOR PUSH)

NOTE: Only Run/Execute the Highlighted once's

It's Uploaded...

now refresh and see github.

DO NOT COPY

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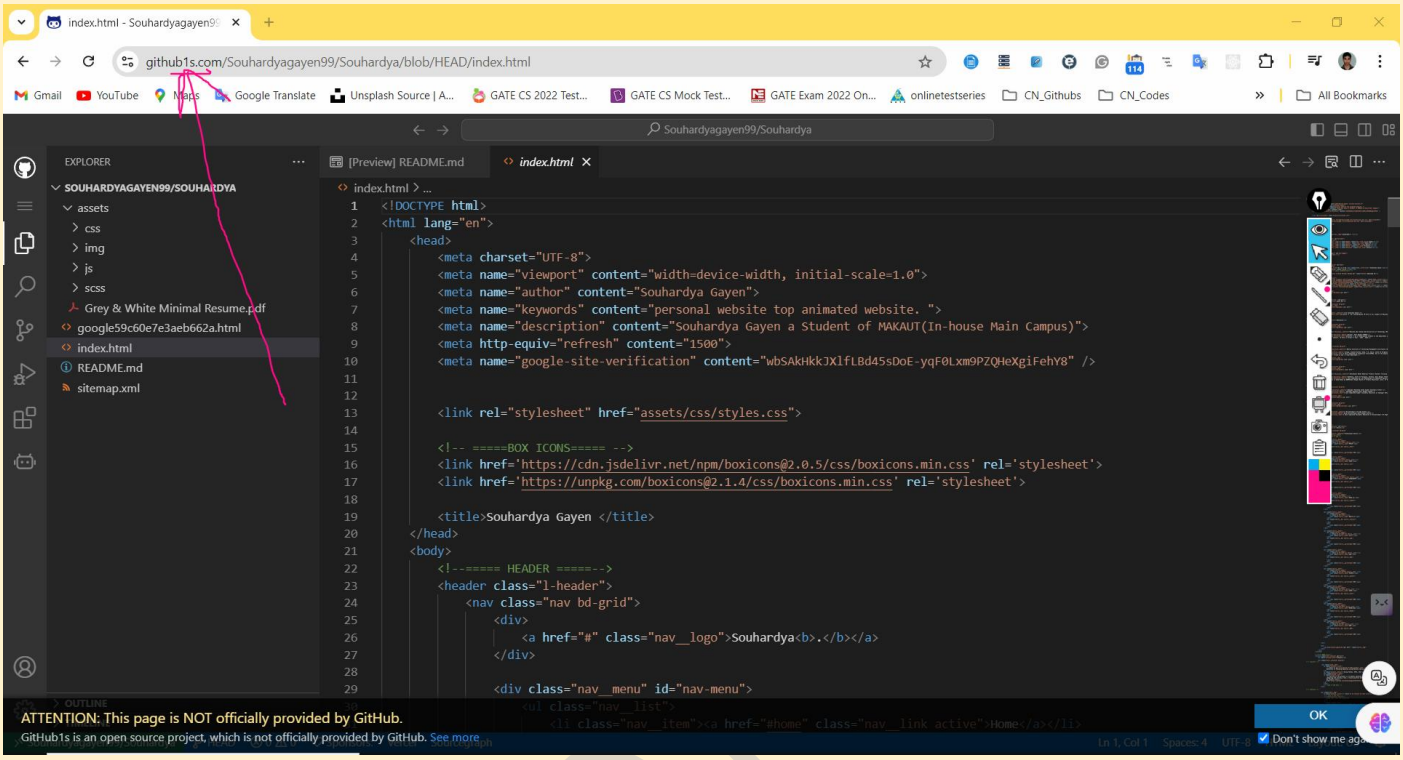
------→ P.T.O

>>TRICK(GitHub1s)

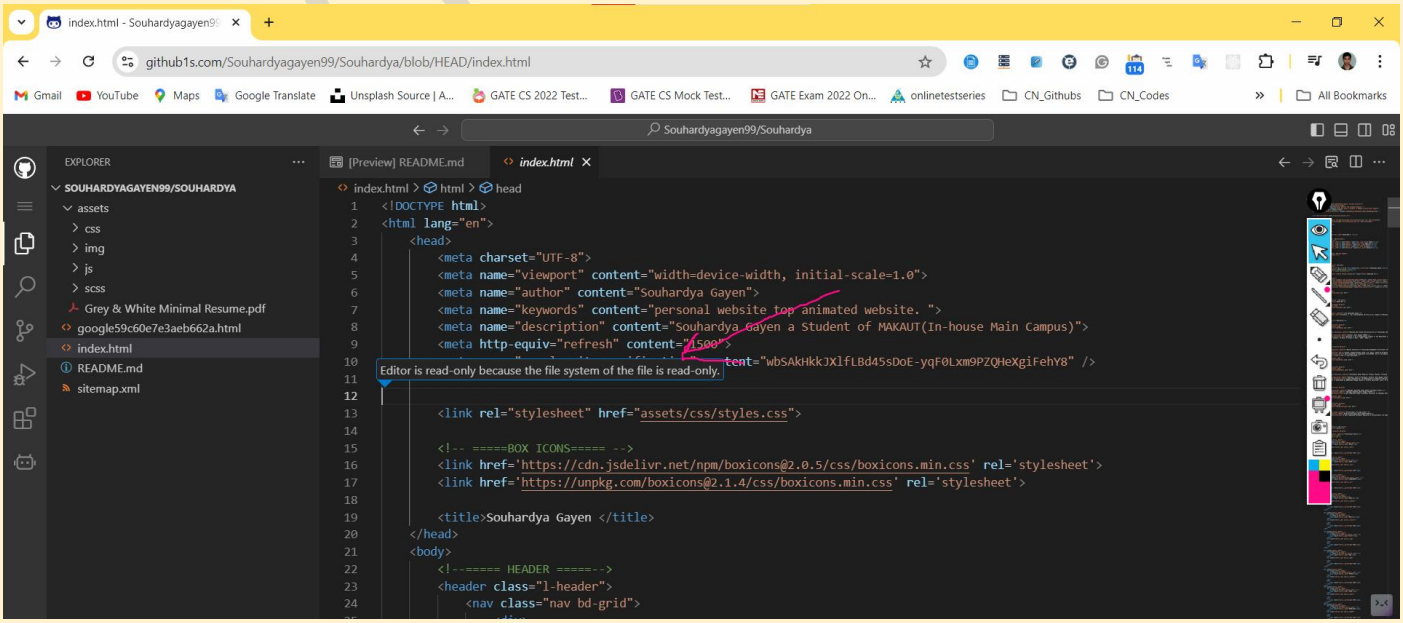
GitHub Repository in VS Code View Directly in Browser /OR, GitHub To VS Code Using GitHub1s:



Choose any of your Repository then Select the address of Github and write '1s' on that address like…

https://github1s.com/Souhardyagayen99/Souhardya/blob/HEAD/index.html

But this is only read only mode we can not write anything here we just see code here like VS Code format.



To Work on Multiple GitHub Branches...

$ git branch -M master //for create & switch on new branch OR,

$ git checkout -b master //for create new branch OR,

$ git branch master //for create new branch

Then add a folder/file in that branch using

$ git add . // (to add all files)

and for commit

$ git commit -m "message here"

and then push that folder/file to that repository using

$ git push origin feature-branch --force

To change the branch on GitHub repository

$ git branch -M master // for create & switch on new branch OR,

$ git checkout master

To merge the branches (we need to have the same no of content & all file name should be same)

$ git merge master // OR,

$ git merge --squash master

For Deleting a Branch

//$ git branch -d master // OR, (it is not working)

$ git push origin --delete master // (used this one)

DO NOT COPY

git pull: This command is a combination of two Git commands: git fetch and git merge.

$ git pull origin main

A lot of things are here to explore….

GitHub will be Continue…..

DO NOT COPY

------→ P.T.O

DO NOT COPY

DO NOT COPY

DO NOT COPY

DO NOT COPY